

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0598

Lutesville 345 kV
April 25, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0598. The Interconnection Customer proposes a 452.8 MW natural gas combined cycle facility interconnecting at the Lutesville 345 kV substation in the Pennsylvania area. The request was received on March 13, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 302 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,351,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
West Campus 345-138 kV XFMR	West Campus-Edes Tap 2 345 KV line	141.9	342	7.1
Big Hollow Rd-West Campus 138 KV line	Edes Tap 2-W Campus 345 KV	123.9	413	7.1
Livermore Falls-Lutesville 138 KV	Edes Tap 2-W Campus 345 KV line	123.2	294	6.0
Sfpp Lp-Grinding 138 KV	Edes Tap 2 Substation-W Campus 345 KV	109.3	254	5.4
Grinding-Strahan 138 KV	Edes Tap 2-W Campus 345 KV	109.3	258	5.4
Edes Ta 2-Strahan 138 KV	Edes Tap 2-W Campus 345 KV Line	108.6	263	5.4
Bluebonnet-Jones Chapel 230 KV line	Robison Park-Russell Fork 230 KV	100.1	894	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
West Campus 345-138 kV XFMR	Replace transformer with higher-rated unit	\$17,830,000	87% (\$15,546,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
West Campus-Big Hollow Road 138 KV	Rebuild line with higher-capacity conductor	\$2,054,000	87% (\$1,791,000)
Livermore Falls-Lutesville 138 KV	Rebuild line with higher-capacity conductor	\$10,272,000	76% (\$7,773,000)
Sfpp Lp-Grinding 138 KV Line	Reconductor line	\$1,715,000	79% (\$1,358,000)
Grinding-Strahan 138 KV	Reconductor line	\$1,323,000	79% (\$1,048,000)
Edes Tap 2 Substation-Strahan 138 KV	Reconductor line	\$11,844,000	79% (\$9,382,000)
Bluebonnet-Jones Chapel 230 KV line	Reconductor line	\$24,263,000	6% (\$1,453,000)
	Total allocated to project		\$38,351,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.